Test 2 is Friday. Today: ten problem types, once each.

- **Turn in Quiz 9** now.
- Test 2 has **10 questions, 20 points each**: about 30% Section 4.2, 20% Section 4.3 and 50% geometry.
- For each type: I work a model, then you try one. Answers appear when we check.
- Most questions have two parts, such as setup and answer. Show your work in each part.

Nothing is printed on Test 2.

Write every formula from memory, including:

- $A = P(1 + r/n)^{nt}$ and $A = Pe^{rt}$
- $\log_b x = y \Leftrightarrow b^y = x$
- Heron: $s = (a + b + c)/2$, $A = \sqrt{(s(s-a)(s-b)(s-c))}$
- radians = degrees × $\pi/180$; $s = r\theta$; $A = \frac{1}{2}r^2\theta$
- $v = r\omega$, with ω = rpm × 2π

$A = P(1 + r/n)^{nt}$

r as a decimal · n = 2 semiannually, 4 quarterly, 12 monthly · round to the nearest cent at the end

Model: $2,500 at 3.5% compounded monthly for 6 years

Setup: $A = 2500(1 + 0.035/12)^{12 \cdot 6} = 2500(1 + 0.035/12)^{72}$

$A \approx \$3{,}083.25$

You try: $7,000 at 5.2% compounded quarterly for 9 years

Setup: $A = 7000(1 + 0.052/4)^{36}$

$A \approx \$11{,}143.92$

$A = Pe^{rt}$

"Compounded continuously" means use e · multiply r × t first · round to the nearest cent

Model: $1,500 at 6.2% compounded continuously for 7 years

Setup: $A = 1500e^{0.062 \cdot 7} = 1500e^{0.434}$

$A \approx \$2{,}315.13$

You try: $9,000 at 3.8% compounded continuously for 11 years

Setup: $A = 9000e^{0.418}$

$A \approx \$13{,}670.29$

Evaluate, then growth or decay

$f(t) = a \cdot b^t$ · $b > 1$: growth · $0 < b < 1$: decay

Model

$P(t) = 450(1.15)^t$, find $P(6) \to \approx 1040.88$; $1.15 > 1$, growth

$f(t) = 3000(0.92)^t$, find $f(4) \to \approx 2149.18$; $0.92 < 1$, decay

You try

$Q(t) = 1800(0.75)^t$, find $Q(3) \to \approx 759.38$; decay

$R(t) = 250(1.06)^t$, find $R(12) \to \approx 503.05$; growth

The log is the exponent

$$b^y = x \iff \log_b x = y$$

Model

(a) $5^3 = 125$ in log form → $\log_5 125 = 3$

(b) Evaluate $\log_5 625$ → 5 to what power is 625? $5^4 = 625$, so 4

You try

(a) $7^2 = 49$ in log form → $\log_7 49 = 2$

(b) Evaluate $\log_9 729$ → $9^3 = 729$, so 3

$b^x = c$: take the log of both sides

$x \cdot \ln b = \ln c \;\rightarrow\; x = \ln c \div \ln b$ · four decimal places

Model: $3^x = 50$

Setup: $x \ln 3 = \ln 50$

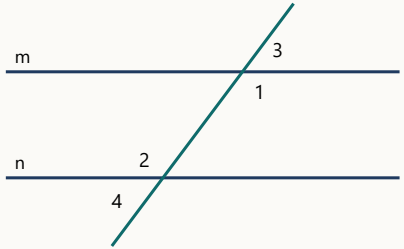
$x = \ln 50 \div \ln 3 \approx 3.5609$

You try: $6^x = 150$

Setup: $x \ln 6 = \ln 150$

$x \approx 2.7965$

Alternate angles are equal



m ∥ n. Angles 1 and 2 are alternate interior; angles 3 and 4 are alternate exterior.

Model: alternate interior angles (2x + 27)° and (4x − 13)°

Equation: $2x + 27 = 4x - 13 \rightarrow x = 20 \rightarrow \text{angle} = 2(20) + 27 = 67°$

You try: alternate exterior angles (5x − 12)° and (3x + 30)°

$5x - 12 = 3x + 30 \rightarrow x = 21 \rightarrow \text{angle} = 93°$

Matching sides are proportional

△ABC ~ △DEF means AB ↔ DE, BC ↔ EF, AC ↔ DF. Keep one triangle on top: AB/BC = DE/EF.

Model: AB = 6, BC = 10, DE = 9, EF = x

Proportion: $6/10 = 9/x$

$6x = 90 \rightarrow x = 15$

You try: AB = 14, BC = 21, DE = 12, EF = x

$14/21 = 12/x$

$14x = 252 \rightarrow x = 18$

Semiperimeter first, then the area

$s = (a + b + c)/2 \cdot A = \sqrt{(s(s - a)(s - b)(s - c))}$

Model: sides 5, 12, 13

$$s = (5 + 12 + 13)/2 = 15$$

$$A = \sqrt{(15 \cdot 10 \cdot 3 \cdot 2)} = \sqrt{900} = 30$$

You try: sides 7, 15, 20

$$s = 21$$

$$A = \sqrt{(21 \cdot 14 \cdot 6 \cdot 1)} = \sqrt{1764} = 42$$

Convert to radians, then s = rθ

$\text{radians} = \text{degrees} \times \pi/180$ · $s = r\theta$ (θ in radians)

Model: r = 8, θ = 120°

$120 \times \pi/180 = 2\pi/3$ rad

$s = 8(2\pi/3) = 16\pi/3 \approx 16.76$

You try: r = 10, θ = 240°

$4\pi/3$ rad

$s = 40\pi/3 \approx 41.89$

$A = \frac{1}{2}r^2\theta$, and $v = r\omega$

θ in radians · ω = rpm × 2π rad per minute · $v = r\omega$ in ft/min

Model: sector r = 10, $\theta = 2\pi/5$; wheel r = 2.5 ft at 160 rpm

$A = \frac{1}{2}(100)(2\pi/5) = 20\pi \approx 62.83$ square units

$\omega = 160(2\pi) = 320\pi$; $v = 2.5(320\pi) = 800\pi \approx 2513.3$ ft/min

You try: sector r = 6, $\theta = 5\pi/4$; wheel r = 1.5 ft at 210 rpm

$A = 22.5\pi \approx 70.69$ square units

$v = 630\pi \approx 1979.2$ ft/min

Where points are usually lost

- Rate as a percent (5.2) instead of a decimal (0.052).
- Wrong n: quarterly is 4, semiannually is 2, monthly is 12.
- Calling a model growth because the starting value is large. Only the base decides.
- Moving the base to the wrong side in log form.
- Rounding in the middle of a calculation.
- Solving for x but not finding the angle.
- Mismatched sides in a similar-triangle proportion.
- Skipping the semiperimeter, or a negative factor in Heron.
- Using degrees in $s = r\theta$ or $A = \frac{1}{2}r^2\theta$.
- Leaving ω in rpm, or using a diameter as the radius.

Before Friday

Tonight and Thursday

- Write all the formulas from memory until you can do it without looking.
- Redo every "you try" from today without notes.
- Work the Test 2 practice PDF on the course page, then check it.
- Take the Test 2 readiness check on the review page until you reach 80%.

Friday, October 30

- Test 2, in class on paper: 10 questions, 200 points, 50 minutes
- No formula sheet: everything is from memory
- Scientific calculator
- Show your work inside each question's box. A correct answer with no work gets no credit.